# Prior-sensitivity diagnostics detect a weak likelihood, not a wrong answer

Operating characteristics in a population-adjusted network, answering part of catalog problem CMU-02

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## Abstract

A proper prior yields a proper posterior even along directions a likelihood cannot identify, so a narrow credible interval in a population-adjusted network can reflect the prior rather than the evidence, and a clean $\hat{R}$ does not rule it out. Catalog entry CMU-02 says the diagnostics that would expose this already exist and have never been calibrated here.

We calibrate them. In a conjugate Gaussian network where the posterior and every power-scaled posterior are exact, so that no diagnostic is contaminated by Monte Carlo error, we measure sensitivity and false-warning rate for prior-to-posterior contraction, a prior-only benchmark, power-scaling sensitivity, a tight-and-loose prior refit, and a composite of the four, against a reference defined by consequence: whether the 95% credible interval misses the truth.

**At the thresholds the literature suggests, they fail.** Intervals missed on 56,210 of 480,000 replicate-contrasts. Pairing each warning with whether that replicate's interval missed, the composite has sensitivity 0.366 (Monte Carlo SE 0.002) at a false-alarm rate of 0.158. Outside two engineered geometries where the likelihood was deliberately made uninformative, sensitivity falls to 0.127. The only rule with useful sensitivity, refitting under halved and doubled prior scales, fires on 53% of the replicates whose intervals were fine.

**The statistics are not the problem; the thresholds are, and something else is.** Used as continuous scores rather than as rules, the same quantities discriminate moderately: area under the curve 0.511 to 0.781 for predicting a miss. So a better-calibrated threshold would do better, and we do not claim otherwise.

**What no threshold can repair** is that these diagnostics measure the prior's *influence* and the harm here comes from the prior's *location*. With disconnected evidence, a tight prior centered at zero and a true interaction of 0.40, coverage is 0.000 at 100 participants per arm with the composite warning on 0.984 of replicates; at 400 per arm coverage is 0.000 and the warning rate is 0.011. More data lets the posterior contract, which every one of these diagnostics reads as reassurance, while none of them can see that the prior is centered in the wrong place.


## 1 The problem

Bayesian evidence synthesis in a population-adjusted network routinely estimates quantities the data barely identify: an interaction informed only by between-study variation, a contrast across a gap in the network, a component effect appearing in few arms. A proper prior with finite marginal likelihood produces a proper posterior along every one of those directions. The output is formatted identically to an estimate the data determined.

Convergence diagnostics do not help, because they interrogate the sampler rather than the likelihood's contribution. CMU-02 states the position precisely: the diagnostics that would expose the problem exist as general Bayesian workflow tools, power-scaling sensitivity (1) with `priorsense` on CRAN and `multinma`'s prior-versus-posterior comparison, and what is missing is their application to these models. The entry adds the reason it matters: **prior dominance grows with scenario severity, so a single check on one benign dataset says nothing about the cases that matter.**

That is an operating-characteristics question, and this study answers it.

## 2 Design

The protocol was registered before the run at `studies/CMU-02-prior-driven-posteriors/protocol.md`. Reporting follows ADEMP (2).

### 2.1 Why the model is exact rather than sampled

Every piece of evidence reduces without loss to a normal observation of a linear function of the treatment parameters, so with $z \sim N(H\theta, V)$ and $\theta \sim N(m_0, S_0)$,

$$S = (H^\top V^{-1} H + S_0^{-1})^{-1}, \qquad m = S(H^\top V^{-1} z + S_0^{-1} m_0). \tag{1}$$

This is deliberate. The study evaluates diagnostics that detect when a posterior is held up by its prior. Were the posterior itself a Monte Carlo approximation, every diagnostic would carry sampling noise of unknown size and a failure to detect could not be separated from a failure to converge. With Equation 1 a failure belongs to the diagnostic. Power-scaling is exact for the same reason: raising a normal prior to the power $\alpha$ multiplies its precision by $\alpha$, so a power-scaled posterior is another closed-form Gaussian rather than an importance-weighted resample.

Parameters are $\theta = (d_B, d_C, d_D, \gamma_B, \gamma_C, \gamma_D)$ with A as reference, where $\gamma_t$ is the change in the conditional treatment effect per standard deviation of the covariate. Between-trial heterogeneity is 0.10, treated as known.

### 2.2 Factors

| Factor | Levels |
|---|---|
| Evidence about treatment C | within-IPD, AgD-wide, AgD-narrow, AgD-flat, disconnected |
| Participants per arm | 100, 400 |
| True $\gamma_C$ | 0, 0.40 |
| Prior scale | tight, regular, weak |
| Target population mean covariate | 0, 0.75 |

120 scenarios, 2000 replicates, two contrasts: the interaction $\gamma_C$ and the target-population marginal C versus B mean difference a committee would read,

$$\Delta_{CB}(\mu_X) = (d_C - d_B) + (\gamma_C - \gamma_B)\,\mu_X, \tag{2}$$

where $\mu_X$ is a **known superpopulation** mean, not a realized target-sample mean and not an estimate, so no uncertainty in it is propagated. Under the identity link this is exact and needs no Monte Carlo evaluation. The posterior contrast vector is $c = (-1, 1, 0, -\mu_X, \mu_X, 0)$ on $(d_B, d_C, d_D, \gamma_B, \gamma_C, \gamma_D)$.

Numerical values, which an earlier version left to the code: true main effects $d_B = 0.30$, $d_C = 0.40$, $d_D = 0.35$; true interactions $\gamma_B = 0.20$, $\gamma_D = 0.10$, with $\gamma_C$ a factor. Priors are independent and zero-centered with standard deviations $(0.20, 0.10)$ for main effects and interactions under **tight**, $(0.50, 0.25)$ under **regular** and $(2.00, 1.00)$ under **weak**. Individual-data trials contribute the exact ordinary-least-squares covariance of their treatment and treatment-by-covariate coefficients for a 1:1 randomized design with the covariate standard normal within trial; aggregate arms contribute a single adjusted contrast with variance $2/n + \tau^2$. The evidence matrices $H$ and $V$ for each geometry are constructed in `R/01-model.R`.

**Prior-truth separation matters for reading the results and is stated here rather than left implicit.** A true $\gamma_C$ of 0.40 sits 4.0 tight-prior standard deviations from the prior mean, 1.6 regular ones and 0.4 weak ones. Under the tight prior in a direction the likelihood barely informs, poor coverage follows arithmetically; peer review was right to ask for this number, and the sample-size result in Section 3.2 should be read as demonstrating a mechanism rather than estimating how often it occurs.

**AgD-flat and AgD-narrow are positive controls**, engineered so the likelihood carries almost no information about the interaction. A diagnostic that fails there has failed at its easiest task; one that fires there has done nothing impressive. Neither is evidence that such structures are common, and results are reported with and without them.

$\gamma_C = 0$ matters because the zero-centered prior is then accidentally correct: a prior-driven analysis is right for the wrong reason, and silence must not be credited.

## 2.3 The reference standard is a consequence, not a geometry

The design as first proposed defined "prior-driven" by the share of posterior variance in directions where likelihood precision is at most a quarter of prior precision, and evaluated the diagnostics against it. That is circular: for a contrast aligned with one eigen-direction, contraction is exactly $\lambda/(1 + \lambda)$, so the reference cut $\lambda \leq 0.25$ **is** the diagnostic cut contraction $< 0.20$. Sensitivity of contraction against it would have been an algebraic identity reported as diagnostic accuracy. Review of the design caught this.

The reference is therefore what goes wrong: a 95% credible interval that misses the truth. A scenario is **harmful** when coverage falls below 0.90. No diagnostic's definition determines that.

## 2.4 Diagnostics

| Diagnostic | Fires when |
|---|---|
| Contraction | $1 - \mathrm{Var}_{\mathrm{post}}/\mathrm{Var}_{\mathrm{prior}} < 0.20$ |
| Prior-only benchmark | **squared** Hellinger distance to the prior $< 0.10$ and decision-probability change $< 0.05$ |
| Power-scaling | prior sensitivity $\geq 0.05$ and likelihood sensitivity $< 0.05$ |
| Tight and loose refit | posterior mean moves $> 0.25$ posterior SD, or decision probability by $> 0.05$ |
| **Composite** | at least two of the four fire |

Power-scaling is measured **distributionally**, as the (unsquared) Hellinger distance between base and power-scaled posteriors per unit $\log \alpha$. The prior-only benchmark uses the **squared** Hellinger distance, as the protocol specifies; the two are on different scales and an earlier draft described both as "Hellinger distance", which peer review flagged. A first implementation used the shift in the posterior *mean*, and that cannot work: in a conjugate Gaussian with a zero-centered prior, raising the prior to $\alpha$ and lowering the likelihood to $1/\alpha$ give posteriors with identical means, so prior and likelihood sensitivity are equal by construction and a rule of the form "prior sensitive, likelihood insensitive" can never fire. Measured that way the diagnostic had sensitivity 0.000 everywhere, which would have been published as a property of the method rather than of our code. The standard deviations do differ, which is why (1) define the diagnostic as a divergence between the two posteriors. They use cumulative Jensen-Shannon distance; a numerical threshold does not transfer between that and Hellinger merely because both are bounded, so the 0.05 used here should be read as a choice informed by their default rather than as their threshold.

The bare posterior is labelled the **no-diagnostic baseline**, not "current practice": `multinma` ships a prior-versus-posterior plot, so what is measured here is what an *automated* warning could achieve, not what a careful analyst reviewing plots would.

# 3 Results

Intervals missed on 56,210 of 480,000 replicate-contrasts. 34 of 240 scenario-contrasts are harmful by the prespecified definition, 18 of them outside the engineered positive controls, so the gate against estimating sensitivity only where failure was manufactured is passed.

**Operating characteristics are computed per replicate**, pairing each warning with whether that replicate's own interval missed. An earlier version called a scenario detected when a rule fired on a majority of its replicates and averaged over scenarios; peer review established that this is not sensitivity for flagging analyses whose intervals miss, that it discards the within-replicate pairing, and that its uncertainty treated fixed factorial design points as a binomial sample. Replicates are Monte Carlo draws, so both the pairing and the Monte Carlo error are well defined at that level.

Table 3: Operating characteristics per replicate. Sensitivity is the share of replicates whose interval missed on which the rule fired; the false-alarm rate is the share of replicates whose interval covered on which it fired. Monte Carlo standard errors in brackets.

| Rule | Sensitivity | False alarm | Sensitivity, controls excluded | False alarm, controls excluded |
|---|---|---|---|---|
| Contraction | 0.363 (0.002) | 0.141 (0.001) | 0.122 | 0.016 |
| Prior-only benchmark | 0.209 (0.002) | 0.128 (0.001) | 0.017 | 0.024 |
| Power-scaling | 0.209 (0.002) | 0.130 (0.001) | 0.007 | 0.003 |
| Tight and loose refit | 0.698 (0.002) | 0.526 (0.001) | 0.830 | 0.531 |
| Composite (2 of 4) | 0.366 (0.002) | 0.158 (0.001) | 0.127 | 0.025 |

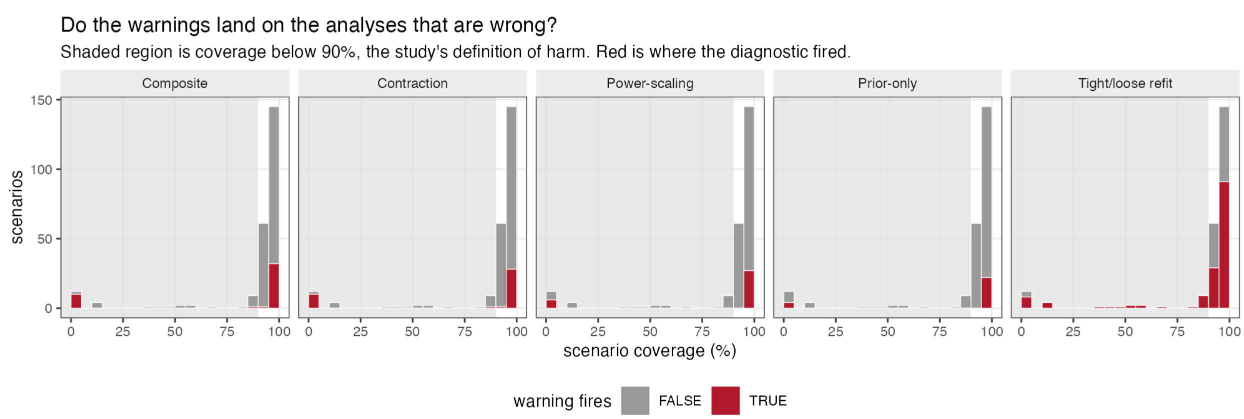


Figure 1: Do the warnings land on the analyses that are wrong?

The prespecified verdict is **diagnostics fail at the thresholds the literature suggests**. Two patterns produce it.

**The sensitive rule is not specific.** Refitting under halved and doubled prior scales catches 0.830 of misses outside the controls and fires on 53% of covered replicates too. A warning attached to half of all sound analyses is not actionable; it is a warning that the model has a prior.

**The specific rules are not sensitive.** Contraction, the prior-only benchmark and power-scaling keep false alarms low and catch almost nothing outside the engineered geometries. Requiring two of four to agree, intended to buy specificity, inherits the low sensitivity instead of repairing it.

### 3.1 The statistics discriminate; the thresholds do not

Table 4: Threshold-free discrimination: area under the curve for predicting that a replicate's interval missed, using each statistic as a continuous score.

| Statistic | AUC, all scenarios | AUC, controls excluded |
|---|---|---|
| contraction | 0.711 | 0.749 |
| prior_sens | 0.766 | 0.781 |
| lik_sens | 0.505 | 0.627 |
| h2 | 0.578 | 0.511 |
| refit_sd | 0.653 | 0.762 |

This matters for what may be concluded. Used as continuous scores the same quantities reach areas under the curve of 0.511 to 0.781 outside the controls, which is moderate discrimination, not none. **The failure reported above is a failure of the prespecified thresholds, not a demonstration that the statistics are uninformative**, and a better-calibrated rule would do better. We do not claim otherwise, and an earlier draft that called the failure structural on the strength of one operating point per rule overstated it.

### 3.2 What no threshold repairs

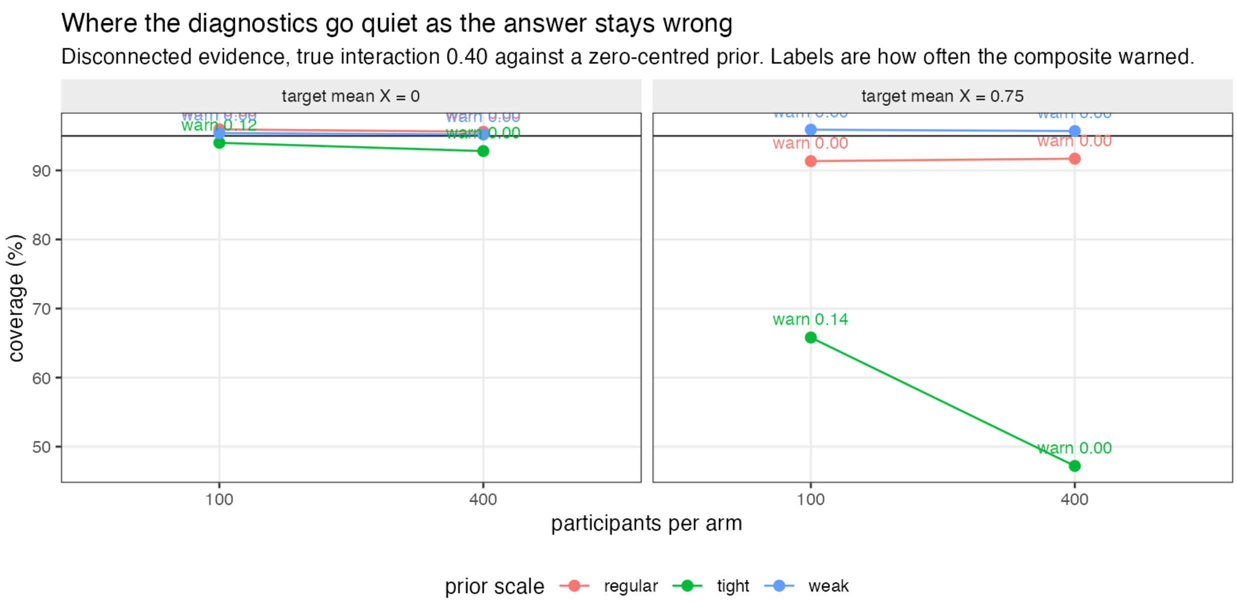


Figure 2: Where the diagnostics go quiet while the answer stays wrong.

Every diagnostic here asks a version of *is the likelihood weak relative to the prior*. The harm asks *is the answer wrong*, and the two separate exactly where it matters most.

Take disconnected evidence with a tight prior centered at zero and a true interaction of 0.40. At 100 participants per arm, coverage of the interaction is 0.000 and the composite warns on 0.984 of replicates. At 400 per arm, coverage is 0.000 and the warning rate is 0.011.

**More data makes the warning quieter while leaving the answer wrong.** The extra data lets the posterior contract, which contraction, the prior-only benchmark and power-scaling all read as reassurance. None of them compares the prior to anything outside itself, so none can see that it is centered in the wrong place. This is the sense in which a prior can only be understood alongside its likelihood (3): a diagnostic that examines the prior's *influence* is silent about the prior's *location*, and no recalibration of a threshold on an influence statistic changes that.

The distinction is not new and the manuscript should not imply it is. Prior-data conflict checking (4) and conflict diagnostics for evidence-synthesis graphs (5) address prior *location* directly, by asking whether the prior and the likelihood are compatible rather than how much the prior contributes. What this study adds is the measurement that the influence-based family, which is what `priorsense` and the prior-versus-posterior plot supply and what CMU-02 names, does not substitute for a conflict check in these networks. A conflict diagnostic is a different instrument and is not evaluated here.

Where the prior is accidentally correct, at $\gamma_C = 0$, the composite fires on 0.190 of covered replicates against 0.119 where it is wrong. Those firings are not errors in the diagnostic's own terms, since the prior really is doing the work; they are counted as false alarms here because the reference is harm, and that mismatch is a real limitation of the reference rather than of the diagnostic.

### 3.3 Measures that were registered and are now reported

An earlier draft omitted several prespecified outputs, which peer review identified.

**The structural rank screen** fires on 0.150 of replicates overall, 0.175 among misses and 0.147 among covered. It fires only where a contrast genuinely lies outside the row space of the evidence, which happens only in the AgD-flat geometry, and it never fires elsewhere. It is the one rule here with no false alarms at all, and it is also the one that answers the narrowest question: exact nonidentification, not weak identification. It cannot see the disconnected failures above, where the contrast is estimable and the answer is still wrong.

**Confident decisions on the wrong side of zero** occur on 0.016 of replicates overall and 0.085 among those whose interval missed.

**The composite adds almost nothing over its best component.** Its sensitivity of 0.366 sits just above contraction alone at 0.363, at a slightly higher false-alarm rate, 0.158 against 0.141. Requiring two of four rules to agree neither buys specificity nor recovers sensitivity here.

**The verdict does not depend on the amended diagnostic.** Power-scaling was reimplemented after a first run, and it contributes to the composite, so the composite recomputed with that component removed entirely gives sensitivity 0.365 and false alarm 0.146, against 0.366 and 0.158 with it.

## 4 What this answers, and what it does not

**Answers, in part, and less than an earlier draft claimed.** CMU-02 asks for these diagnostics to be applied to population-adjusted models and evaluated conditional on scenario severity. What is established here is narrower: in a conjugate Gaussian network with graded evidence structures, these warnings coincide poorly with realized interval misses at the thresholds examined, and the influence-based family cannot see a misplaced prior. That is not the same as calibrating them for detecting prior dominance, which would need a reference for dominance that is not one of the diagnostics; finding such a reference is itself unfinished business, since the obvious geometric one is algebraically identical to contraction. An analyst who runs them and sees nothing has learned that the likelihood is not obviously weak, which is a much smaller claim than that the answer is trustworthy.

**Does not answer.** The reference standard is undercoverage, not prior dominance, and those are not the same thing: a diagnostic that correctly detects a dominant prior in a cell where the prior happens to be right is counted here as a false alarm. That mismatch was chosen deliberately, because the geometric alternative was algebraically identical to one of the diagnostics, but it means these numbers answer "do the warnings predict wrong answers" rather than "do the warnings detect prior dominance".

Only one form of prior misspecification is examined, a zero-centered prior against a nonzero truth, at three scales. A diffuse but miscentered prior is untested, though the argument in Section 3.2 would apply to it too. Only automated thresholds are evaluated, and those thresholds come from the source literature for power-scaling and are our own choices for the others; a plot read by an experienced analyst is a different instrument. The harm threshold of 0.90 coverage is prespecified but its sensitivity to that choice is not explored.

The other half of CMU-02, wiring these into released software, is untouched; neither `multinma` nor `cpaic` is run. The Stan validation of the Gaussian reduction named in the protocol was not run. The model is conjugate Gaussian with an identity link, a correctly specified linear mean and known variances, so nothing here transfers directly to a non-conjugate posterior where the diagnostics behave differently and MCMC error enters. Only automated thresholds are evaluated; a plot read by an experienced analyst is a different instrument and may do better. Thresholds are those the source literature suggests and were not tuned, so a better-calibrated rule may exist, though the structural argument in Section 3.2 suggests re-tuning cannot fix the failure it describes.

It bears on **CMP-14** only for the generic case of an interaction informed by aggregate data alone. CMP-14 concerns component models and this design has treatment-specific parameters, so the component-specific part is untouched. An earlier draft claimed IDN-06 as well; that is withdrawn, because this design contains no within-study versus between-study discrepancy at all.

# 5 Peer review

Reviewed in two rounds by two independent reviewers. Reports, responses and the editorial decision are published in full at `studies/CMU-02-prior-driven-posteriors/review/peer-review.md`.

# 6 Reproducibility

R: R version 4.6.0 (2026-04-24) Platform: aarch64-apple-darwin23 Running under: macOS Tahoe 26.5

## 6.1 Packages

| package | version |
|---|---|
| base | 4.6.0 |
| stats | 4.6.0 |
| Matrix | 1.7.5 |
| future | 1.70.0 |
| furrr | 0.4.0 |

## 6.2 Run

- **study**: CMU-02 prior-driven posteriors
- **scenarios**: 120
- **replicates_per_scenario**: 2000
- **total**: 240000
- **master_seed**: 20260729

```
Rscript R/03-run.R
Rscript R/04-analyze.R
Rscript R/05-figures.R
```

## References


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